Networks and Causal Inference Short Course: Part 1

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Using Data from the Transmission Reduction Intervention Project in Athens, Greece

1. Step 1: Download the data and load it into R Studio

This is a simulated toy dataset made for this course. It is based on data collected from the Transmission Reduction Intervention Project in Athens Greece. Public data here: https://www.icpsr.umich.edu/web/NAHDAP/studies/39059/versions/V2/staff

Read more: https://pubmed.ncbi.nlm.nih.gov/27917890/

```r
# -----------------------------
# 1. Read data into R
# -----------------------------

#install packages (if needed)
#install.packages(c("readxl","dplyr","ggplot2","geepack","tidyverse"))
#load packages
library(readxl)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(knitr)
```

```
## Warning: package 'knitr' was built under R version 4.4.3
```

```r
library(ggplot2)
library(geepack)
```

```
## Warning: package 'geepack' was built under R version 4.4.1
```

```r
library(tidyverse)
```

```
## Warning: package 'tibble' was built under R version 4.4.3

## Warning: package 'purrr' was built under R version 4.4.3

## Warning: package 'stringr' was built under R version 4.4.1

## -- Attaching core tidyverse packages ----------------------- tidyverse 2.0.0 --
```

```
## v forcats   1.0.0     v stringr   1.6.0
## v lubridate 1.9.3     v tibble    3.3.1
## v purrr     1.2.1     v tidyr     1.3.1
## v readr     2.1.5
```

```
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(cobalt)
```

```
## Warning: package 'cobalt' was built under R version 4.4.3
```

```
##  cobalt (Version 4.6.2, Build Date: 2026-01-29)
```

```r
library(purrr)
library(scales)
```

```
##
## Attaching package: 'scales'
##
## The following object is masked from 'package:purrr':
##
##     discard
##
## The following object is masked from 'package:readr':
##
##     col_factor
```

```r
#will need to test on final folder structure shared - we can post to GitHub
nodes <- read.csv("../TRIP_network_simulation/TRIPsim_nodes.csv")
edges <- read.csv("../TRIP_network_simulation/TRIPsim_edges.csv")

head(nodes)
```

```
##   id num_nn component posneg date share education employment age avg_posneg
## 1  1      4         1      1    1     1         0          1  39  1.0000000
## 2  2      3         1      1    0     0         1          0  25  0.6666667
## 3  3      1         1      1    0     1         0          1  40  1.0000000
## 4  4      3         1      0    1     1         1          0  37  0.6666667
## 5  5      4         1      1    0     1         1          0  28  0.5000000
## 6  6      2         1      1    1     0         0          1  17  1.0000000
##    avg_date avg_share avg_education avg_employment avg_age     raneff treatment
## 1 0.5000000 0.5000000     0.7500000      0.5000000   27.75 -0.1141261         0
## 2 1.0000000 1.0000000     0.6666667      0.6666667   35.67 -0.1141261         0
## 3 0.0000000 1.0000000     0.0000000      1.0000000   31.00 -0.1141261         1
## 4 0.3333333 0.3333333     0.6666667      0.3333333   29.00 -0.1141261         0
## 5 0.7500000 0.5000000     0.5000000      0.2500000   36.75 -0.1141261         0
## 6 1.0000000 0.5000000     0.5000000      0.5000000   38.00 -0.1141261         1
##   num_tnn num_utnn outcome
## 1       0        4       1
## 2       0        3       0
## 3       0        1       1
## 4       0        3       0
## 5       1        3       1
## 6       1        1       0
```

```r
head(edges)
```

```
##   from to
## 1    1  2
## 2    2  4
## 3    5  8
## 4    1  9
## 5    3  9
## 6    1 10
```

Generate descriptive statistical analyses and histograms

```r
# -----------------------------
# 2. Descriptive Analysis
# -----------------------------

# Basic summary statistics
summary(nodes)
```

```
##        id             num_nn       component          posneg      
##  Min.   :  1.00   Min.   :1.00   Min.   : 1.000   Min.   :0.0000  
##  1st Qu.: 54.75   1st Qu.:2.00   1st Qu.: 3.000   1st Qu.:0.0000  
##  Median :108.50   Median :3.00   Median : 5.000   Median :1.0000  
##  Mean   :108.50   Mean   :3.25   Mean   : 6.333   Mean   :0.5278  
##  3rd Qu.:162.25   3rd Qu.:4.00   3rd Qu.: 9.000   3rd Qu.:1.0000  
##  Max.   :216.00   Max.   :8.00   Max.   :20.000   Max.   :1.0000  
##       date            share          education        employment   
##  Min.   :0.0000   Min.   :0.0000   Min.   :0.0000   Min.   :0.000  
##  1st Qu.:0.0000   1st Qu.:1.0000   1st Qu.:0.0000   1st Qu.:0.000  
##  Median :1.0000   Median :1.0000   Median :0.0000   Median :0.000  
##  Mean   :0.5185   Mean   :0.7685   Mean   :0.4167   Mean   :0.463  
##  3rd Qu.:1.0000   3rd Qu.:1.0000   3rd Qu.:1.0000   3rd Qu.:1.000  
##  Max.   :1.0000   Max.   :1.0000   Max.   :1.0000   Max.   :1.000  
##       age          avg_posneg        avg_date        avg_share    
##  Min.   :14.00   Min.   :0.0000   Min.   :0.0000   Min.   :0.000  
##  1st Qu.:29.00   1st Qu.:0.3333   1st Qu.:0.3333   1st Qu.:0.575  
##  Median :35.00   Median :0.5000   Median :0.5000   Median :0.800  
##  Mean   :34.86   Mean   :0.5328   Mean   :0.5319   Mean   :0.767  
##  3rd Qu.:40.00   3rd Qu.:0.7500   3rd Qu.:0.7500   3rd Qu.:1.000  
##  Max.   :58.00   Max.   :1.0000   Max.   :1.0000   Max.   :1.000  
##  avg_education    avg_employment      avg_age          raneff         
##  Min.   :0.0000   Min.   :0.0000   Min.   :19.00   Min.   :-1.096824  
##  1st Qu.:0.0000   1st Qu.:0.2500   1st Qu.:31.25   1st Qu.:-0.194025  
##  Median :0.4143   Median :0.5000   Median :34.71   Median :-0.083398  
##  Mean   :0.4223   Mean   :0.4508   Mean   :34.54   Mean   :-0.004333  
##  3rd Qu.:0.6667   3rd Qu.:0.6667   3rd Qu.:37.50   3rd Qu.: 0.212757  
##  Max.   :1.0000   Max.   :1.0000   Max.   :53.00   Max.   : 0.870289  
##    treatment         num_tnn          num_utnn        outcome      
##  Min.   :0.0000   Min.   :0.0000   Min.   :0.000   Min.   :0.0000  
##  1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:2.000   1st Qu.:0.0000  
##  Median :0.0000   Median :0.0000   Median :3.000   Median :0.0000  
##  Mean   :0.1296   Mean   :0.3565   Mean   :2.894   Mean   :0.4861  
##  3rd Qu.:0.0000   3rd Qu.:1.0000   3rd Qu.:4.000   3rd Qu.:1.0000  
##  Max.   :1.0000   Max.   :3.0000   Max.   :8.000   Max.   :1.0000
```

```r
make_table1 <- function(data, cat_vars, num_vars) {

  # Categorical variables
  cat_table <- map_dfr(cat_vars, function(var) {
    data %>%
      count(.data[[var]]) %>%
      mutate(
        variable = var,
        level = as.character(.data[[var]]),
        stat = paste0(n, " (", percent(n / sum(n), accuracy = 0.1), ")")
      ) %>%
      select(variable, level, stat)
  })

  # Numeric variables
  num_table <- map_dfr(num_vars, function(var) {
    data %>%
      summarise(
        mean = mean(.data[[var]], na.rm = TRUE),
        sd   = sd(.data[[var]], na.rm = TRUE)
      ) %>%
      mutate(
        variable = var,
        level = "",
        stat = sprintf("%.2f (%.2f)", mean, sd)
      ) %>%
      select(variable, level, stat)
  })

  # Combine
  bind_rows(cat_table, num_table)
}

table1 <- make_table1(
  data = nodes,
  cat_vars = c("posneg", "share", "education", "employment", "treatment", "outcome"),
  num_vars = c("num_nn", "num_tnn", "num_utnn")
)

table1
```

```
##      variable level        stat
## 1      posneg     0 102 (47.2%)
## 2      posneg     1 114 (52.8%)
## 3       share     0  50 (23.1%)
## 4       share     1 166 (76.9%)
## 5   education     0 126 (58.3%)
## 6   education     1  90 (41.7%)
## 7  employment     0 116 (53.7%)
## 8  employment     1 100 (46.3%)
## 9   treatment     0 188 (87.0%)
## 10  treatment     1  28 (13.0%)
## 11    outcome     0 111 (51.4%)
## 12    outcome     1 105 (48.6%)
```

```
## 13      num_nn       3.25 (1.55)
## 14     num_tnn       0.36 (0.62)
## 15    num_utnn       2.89 (1.49)
```

```r
# Frequency tables for categorical variables
nodes %>% count(posneg)
```

```
##   posneg   n
## 1      0 102
## 2      1 114
```

```r
nodes %>% count(share)
```

```
##   share   n
## 1     0  50
## 2     1 166
```

```r
nodes %>% count(education)
```

```
##   education   n
## 1         0 126
## 2         1  90
```

```r
nodes %>% count(employment)
```

```
##   employment   n
## 1          0 116
## 2          1 100
```

```r
nodes %>% count(treatment)
```

```
##   treatment   n
## 1         0 188
## 2         1  28
```

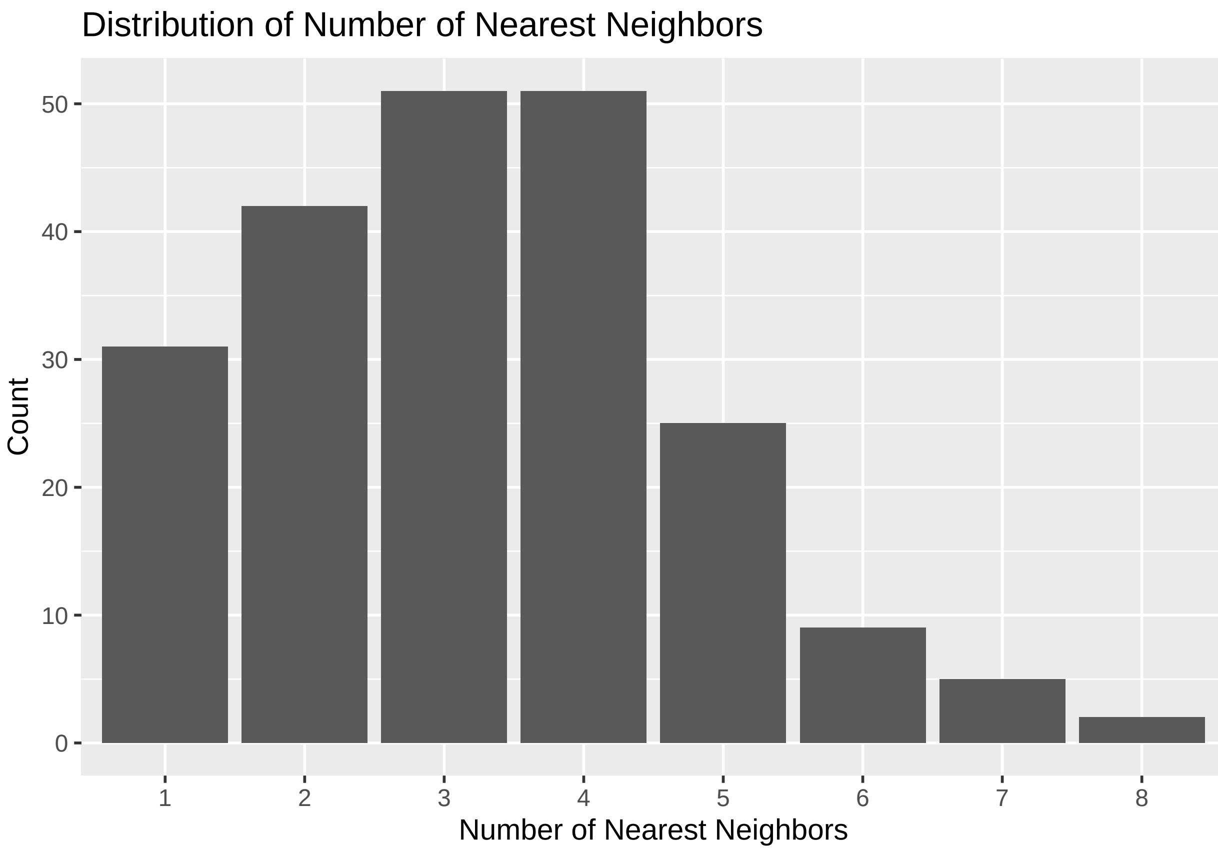
```r
nodes %>% count(outcome)
```

```
##   outcome   n
## 1       0 111
## 2       1 105
```

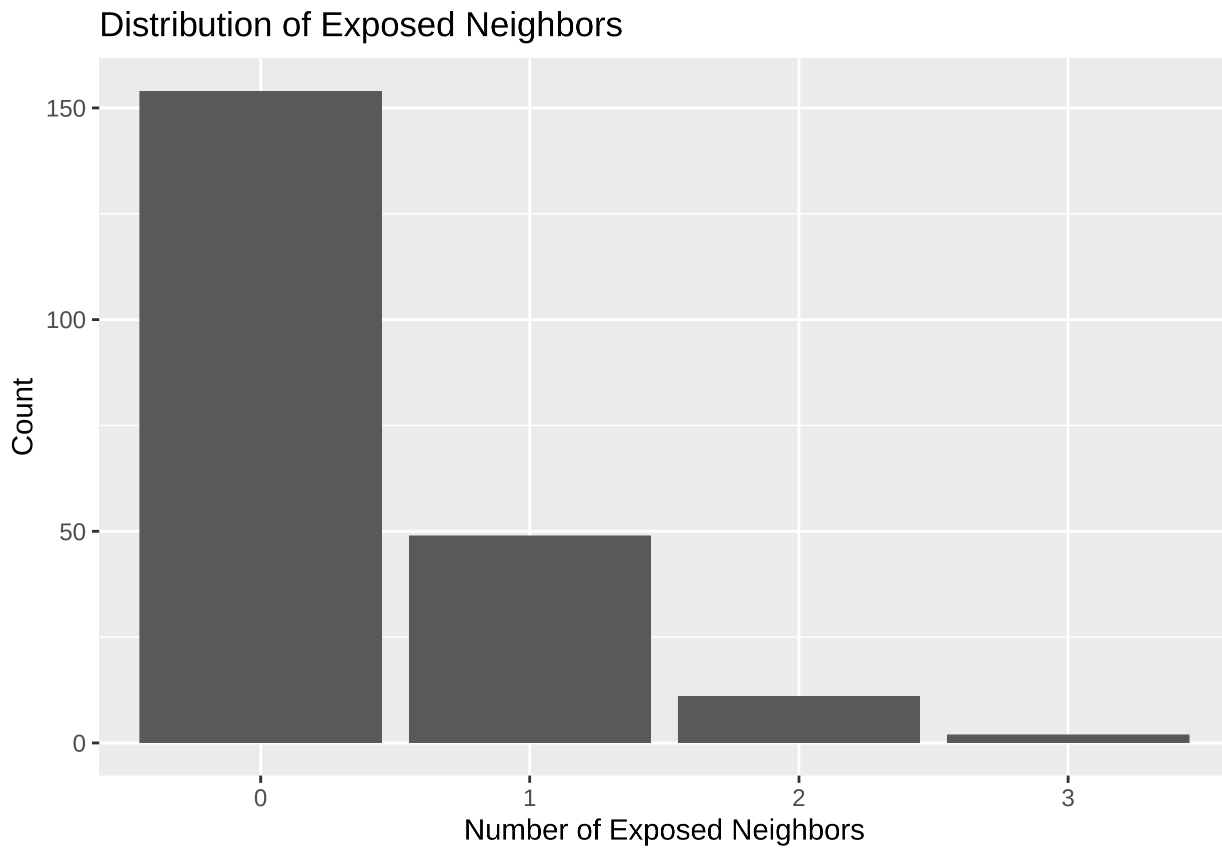
```r
# Means of key numeric variables
nodes %>%
  summarise(
    mean_nn = mean(num_nn, na.rm = TRUE),
    mean_tnn = mean(num_tnn, na.rm = TRUE),
    mean_utnn = mean(num_utnn, na.rm = TRUE)
  )
```

```
##   mean_nn  mean_tnn mean_utnn
## 1    3.25 0.3564815  2.893519
```

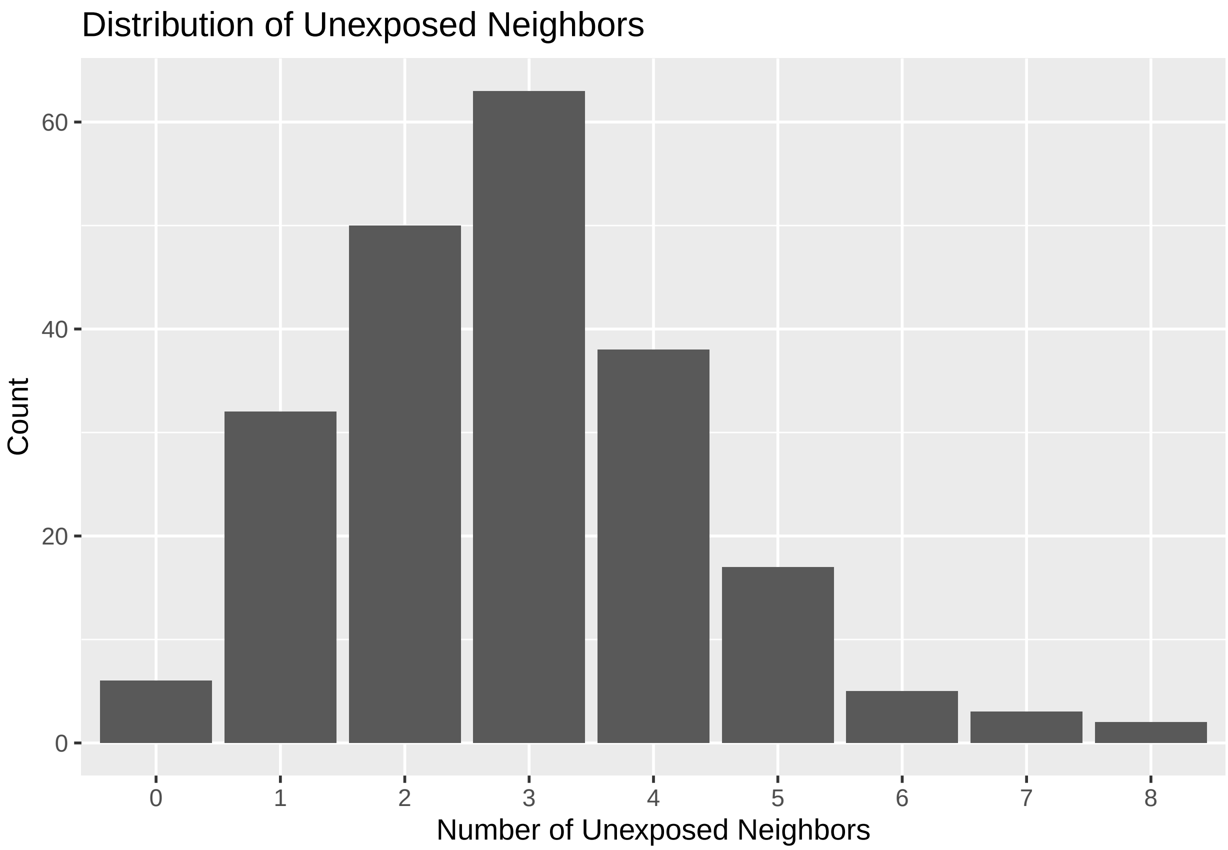
```r
# Plots of network-related variables

ggplot(nodes, aes(x = factor(num_nn))) +
  geom_bar() +
  scale_x_discrete(breaks = 0:9, labels = 0:9) +
  labs(title = "Distribution of Number of Nearest Neighbors",
       x = "Number of Nearest Neighbors",
       y = "Count")
```




```r
ggplot(nodes, aes(x = factor(num_tnn))) +
  geom_bar() +
  scale_x_discrete(breaks = 0:4, labels = 0:4) +
  labs(title = "Distribution of Exposed Neighbors",
       x = "Number of Exposed Neighbors",
       y = "Count")
```




```r
ggplot(nodes, aes(x = factor(num_utnn))) +
  geom_bar() +
  scale_x_discrete(breaks = 0:8, labels = 0:8) +
  labs(title = "Distribution of Unexposed Neighbors",
       x = "Number of Unexposed Neighbors",
       y = "Count")
```



Analysis of the association between community alerts and any risk behavior outcome, unadjusted and adjusted for baseline covariates.

```
# -----------------------------
# 3. Assess association
# -----------------------------

#Conditional RD,RR - condition on covariates in outcome model (not marginal)

#Unadjusted models
#Risk Difference

model_rd_unadj <- glm(
  outcome ~ treatment,
  data = nodes,
  family = binomial(link = "identity")
)

#Risk Ratio
model_rr_unadj <- glm(
  outcome ~ treatment,
  data = nodes,
  family = binomial(link = "log")
)

#adjusted models
```

```r
#Risk Difference
# Binomial Log model did not converge - fit with GEE log-Poisson to get a robust estimator of the varia
#rescale age (per 10 year increase)
nodes$age10 <- nodes$age/10

 # model_rr <- glm(
 #   outcome ~ treatment + age10 + posneg + share + education + employment,
 #   data = nodes,
 #   family = binomial(link = "log")
 # )

gee_rr <- geeglm(
  outcome ~ treatment + age10 + posneg + date + share + education + employment,
  data = nodes,
  id = id,
  family = poisson(link = "log"),
  corstr = "independence"
)

#Risk Ratio
# Binomial Identity model did not converge - fit with GEE Normal-identity to get a robust estimator of
# model_rd <- glm(
#   outcome ~ treatment + posneg + share + education + employment,
#   data = nodes,
#   family = binomial(link = "identity")
# )

gee_rd <- geeglm(
  outcome ~ treatment + age10 + posneg + date + share + education + employment,
  data = nodes,
  id = id,
  family = gaussian(link = "identity"),
  corstr = "independence"
)

#Format output into a table
# Associational RR and RD

# Unadjusted RR
b_rr_unadj <- coef(model_rr_unadj)["treatment"]
se_rr_unadj <- sqrt(vcov(model_rr_unadj)["treatment", "treatment"])
rr_unadj <- exp(b_rr_unadj)
rr_unadj_ci <- exp(b_rr_unadj + c(-1, 1) * 1.96 * se_rr_unadj)

# Unadjusted RD
b_rd_unadj <- coef(model_rd_unadj)["treatment"]
se_rd_unadj <- sqrt(vcov(model_rd_unadj)["treatment", "treatment"])
rd_unadj <- b_rd_unadj
rd_unadj_ci <- b_rd_unadj + c(-1, 1) * 1.96 * se_rd_unadj

# Adjusted RR
b_rr_adj <- coef(gee_rr)["treatment"]
se_rr_adj <- sqrt(vcov(gee_rr)["treatment", "treatment"])
```

```r
rr_adj <- exp(b_rr_adj)
rr_adj_ci <- exp(b_rr_adj + c(-1, 1) * 1.96 * se_rr_adj)

# Adjusted RD
b_rd_adj <- coef(gee_rd)["treatment"]
se_rd_adj <- sqrt(vcov(gee_rd)["treatment", "treatment"])
rd_adj <- b_rd_adj
rd_adj_ci <- b_rd_adj + c(-1, 1) * 1.96 * se_rd_adj


results <- data.frame(
  model = c("Unadjusted", "Unadjusted", "Adjusted", "Adjusted"),
  measure = c("Risk ratio", "Risk difference", "Risk ratio", "Risk difference"),
  estimate = c(rr_unadj, rd_unadj, rr_adj, rd_adj),
  lower_95_ci = c(rr_unadj_ci[1], rd_unadj_ci[1], rr_adj_ci[1], rd_adj_ci[1]),
  upper_95_ci = c(rr_unadj_ci[2], rd_unadj_ci[2], rr_adj_ci[2], rd_adj_ci[2])
)

results
```

```
##        model         measure     estimate lower_95_ci upper_95_ci
## 1 Unadjusted      Risk ratio  1.032967033   0.6933062   1.5390327
## 2 Unadjusted Risk difference  0.015957447  -0.1825452   0.2144601
## 3   Adjusted      Risk ratio  1.024567106   0.7286820   1.4405979
## 4   Adjusted Risk difference -0.001150168  -0.1754597   0.1731593
```

```r
#Round to 3 decimal places
results$estimate <- round(results$estimate, 3)
results$lower_95_ci <- round(results$lower_95_ci, 3)
results$upper_95_ci <- round(results$upper_95_ci, 3)

results
```

```
##        model         measure estimate lower_95_ci upper_95_ci
## 1 Unadjusted      Risk ratio    1.033       0.693       1.539
## 2 Unadjusted Risk difference    0.016      -0.183       0.214
## 3   Adjusted      Risk ratio    1.025       0.729       1.441
## 4   Adjusted Risk difference   -0.001      -0.175       0.173
```

Check the positivity assumption for random violations, making tables of the exposure to the community alerts by each covariate. Based on the exposure, there are no structural violations of positivity (exposure possible for all combinations of individual and group level covariates).

```r
# ----------------------------------------
# 5. Assess random violations of positivity
# ----------------------------------------
# Table Function
make_combined_table <- function(data, treatment_var, covariates, digits = 2) {

  treatment_var <- rlang::ensym(treatment_var)

  # Pivot all covariates into long format
  long_data <- data %>%
    pivot_longer(
      cols = all_of(covariates),
```

```r
      names_to = "variable",
      values_to = "level"
    ) %>%
    mutate(level = as.character(level))

  # Count by treatment + totals
  tab <- long_data %>%
    count(variable, level, !!treatment_var) %>%
    pivot_wider(
      names_from = !!treatment_var,
      values_from = n,
      values_fill = 0
    )

  # Rename treatment columns (assumes 0/1)
  tab <- tab %>%
    rename(
      Untreated = `0`,
      Treated   = `1`
    )

  # Add totals
  tab <- tab %>%
    mutate(
      Total = Untreated + Treated
    )

  # percentages
  tab <- tab %>%
    group_by(variable) %>%
    mutate(
      Untreated = paste0(Untreated, " (", round(Untreated / sum(Untreated) * 100, digits), "%)"),
      Treated   = paste0(Treated, " (", round(Treated / sum(Treated) * 100, digits), "%)"),
      Total     = paste0(Total, " (", round(Total / sum(Total) * 100, digits), "%)")
    ) %>%
    ungroup()

  tab %>%
    arrange(variable, level)
}

table_combined <- make_combined_table(
  data = nodes,
  treatment_var = treatment,
  covariates = c(
    "posneg", "share", "education", "employment", "date"
  )
)

table_combined
```

```
## # A tibble: 10 x 5
##    variable   level Untreated    Treated     Total
##    <chr>      <chr> <chr>        <chr>       <chr>
```

```
## 1 date       0    90 (47.87%)  14 (50%)    104 (48.15%)
## 2 date       1    98 (52.13%)  14 (50%)    112 (51.85%)
## 3 education  0    111 (59.04%) 15 (53.57%) 126 (58.33%)
## 4 education  1    77 (40.96%)  13 (46.43%) 90 (41.67%)
## 5 employment 0    105 (55.85%) 11 (39.29%) 116 (53.7%)
## 6 employment 1    83 (44.15%)  17 (60.71%) 100 (46.3%)
## 7 posneg     0    88 (46.81%)  14 (50%)    102 (47.22%)
## 8 posneg     1    100 (53.19%) 14 (50%)    114 (52.78%)
## 9 share      0    39 (20.74%)  11 (39.29%) 50 (23.15%)
## 10 share     1    149 (79.26%) 17 (60.71%) 166 (76.85%)
```

```r
make_continuous_table <- function(data, treatment_var, cont_vars, digits = 2, show_sd = TRUE) {

  treatment_var <- rlang::ensym(treatment_var)

  map_dfr(cont_vars, function(var) {

    var_sym <- rlang::sym(var)

    # Means
    untreated_mean <- data %>%
      filter(!!treatment_var == 0) %>%
      summarise(m = mean(!!var_sym, na.rm = TRUE)) %>%
      pull(m)

    treated_mean <- data %>%
      filter(!!treatment_var == 1) %>%
      summarise(m = mean(!!var_sym, na.rm = TRUE)) %>%
      pull(m)

    overall_mean <- mean(data[[var]], na.rm = TRUE)

    if (show_sd) {
      untreated_sd <- data %>%
        filter(!!treatment_var == 0) %>%
        summarise(s = sd(!!var_sym, na.rm = TRUE)) %>%
        pull(s)

      treated_sd <- data %>%
        filter(!!treatment_var == 1) %>%
        summarise(s = sd(!!var_sym, na.rm = TRUE)) %>%
        pull(s)

      overall_sd <- sd(data[[var]], na.rm = TRUE)

      tibble(
        Variable = var,
        Untreated = sprintf("%.2f (%.2f)", untreated_mean, untreated_sd),
        Treated   = sprintf("%.2f (%.2f)", treated_mean, treated_sd),
        Total     = sprintf("%.2f (%.2f)", overall_mean, overall_sd)
      )

    } else {
```

```r
      tibble(
        Variable = var,
        Untreated = round(untreated_mean, digits),
        Treated   = round(treated_mean, digits),
        Total     = round(overall_mean, digits)
      )
    }
  })
}

table_continuous <- make_continuous_table(
  data = nodes,
  treatment_var = treatment,
  cont_vars = c("avg_posneg", "avg_share", "avg_education", "avg_employment", "avg_date")
)

table_continuous
```

```
## # A tibble: 5 x 4
##   Variable       Untreated   Treated     Total
##   <chr>          <chr>       <chr>       <chr>
## 1 avg_posneg     0.52 (0.33) 0.65 (0.37) 0.53 (0.33)
## 2 avg_share      0.77 (0.25) 0.77 (0.30) 0.77 (0.26)
## 3 avg_education  0.42 (0.32) 0.45 (0.37) 0.42 (0.33)
## 4 avg_employment 0.46 (0.30) 0.41 (0.35) 0.45 (0.31)
## 5 avg_date       0.51 (0.32) 0.67 (0.36) 0.53 (0.33)
```

Estimation using Inverse Probability Weighted Estimator (propensity score model)

```r
# -----------------------------
# 5. IPW Estimator
# -----------------------------

#Fit treatment model
mylogit <- glm(treatment ~ age10 + posneg + date + share + education + employment , data = nodes, famil

#propensity score denominator
denom.p <- predict(mylogit,type = "response")

#stabilized weights
#numerator
numer.fit=glm(treatment~1,family=binomial(), data=nodes)
numer.p=predict(numer.fit,type="response")

#denominator

#unstabilized weights
nodes$w =ifelse(nodes$treatment==0,(1-denom.p),denom.p)
#stabilized weights
nodes$sw=ifelse(nodes$treatment==0,((1-numer.p)/(1-denom.p)),(numer.p/denom.p))

#check distribution of stabilized weights
#Mean around 1
#Deviation from 1 means possible model mispecification or positivity issue
summary(nodes$sw)
```

```
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.3206 0.9322 0.9740 1.0027 1.0340 2.8083
```

```r
# IPW outcome model (marginal structural model) with robust variance
# Causal Risk Ratio
# Fit with stabilized weights
gee_rr_ipw <- geeglm(
  outcome ~ treatment,
  weights = sw,
  data = nodes,
  id = id,
  family = binomial(link = "log"),
  corstr = "independence"
)
```

```
## Warning in eval(family$initialize): non-integer #successes in a binomial glm!
```

```r
#Warning in eval(family$initialize) :
  #non-integer #successes in a binomial glm!
  #Warning OK because we weighted geeglm

b_rr <- coef(gee_rr_ipw)["treatment"]
se_rr <- sqrt(vcov(gee_rr_ipw)["treatment", "treatment"])

rr <- exp(b_rr)
rr_ci <- exp(b_rr + c(-1, 1) * 1.96 * se_rr)

# Causal risk difference
# Fit with stabilized weights
gee_rd_ipw <- geeglm(
  outcome ~ treatment,
  weights = sw,
  data = nodes,
  id = id,
  family = gaussian(link = "identity"),
  corstr = "independence"
)

b_rd <- coef(gee_rd_ipw)["treatment"]
se_rd <- sqrt(vcov(gee_rd_ipw)["treatment", "treatment"])

rd <- b_rd
rd_ci <- b_rd + c(-1, 1) * 1.96 * se_rd

results_ipw <- data.frame(
  measure = c("Risk ratio", "Risk difference"),
  estimate = c(rr, rd),
  lower_95_ci = c(rr_ci[1], rd_ci[1]),
  upper_95_ci = c(rr_ci[2], rd_ci[2])
)

results_ipw$estimate <- round(results_ipw$estimate, 3)
results_ipw$lower_95_ci <- round(results_ipw$lower_95_ci, 3)
results_ipw$upper_95_ci <- round(results_ipw$upper_95_ci, 3)
```

```r
results_ipw
```

```
##           measure estimate lower_95_ci upper_95_ci
## 1      Risk ratio    0.999       0.628       1.589
## 2 Risk difference   -0.001      -0.227       0.226
```

Estimation using the G-computation (outcome model)

```r
# -----------------------------
# 6. G-computation
# -----------------------------

# Function for G-computation
# Save as a function so we can bootstrap to get confidence interval
# Under SUTVA and IID, we sample each individual with replacement for the boot samples

gcomp_est <- function(data) {
  n <- nrow(data)

  # original data
  all <- data
  all$interv <- -1

  # counterfactual copy with A = 0
  all.untr <- all
  all.untr$interv    <- 0
  all.untr$treatment <- 0
  all.untr$outcome  <- NA

  # counterfactual copy with A = 1
  all.tr <- all
  all.tr$interv    <- 1
  all.tr$treatment <- 1
  all.tr$outcome   <- NA

  onesample <- as.data.frame(rbind(all, all.untr, all.tr))

  # outcome model
  # Use Binomial-Logit then take function of probabilities
  glm.obj <- glm(
    outcome ~ treatment + age10 + posneg + date + share + education + employment,
    data = onesample,
    family = binomial(link = "logit")
  )
  summary(glm.obj)

  onesample$meanY <- predict(glm.obj, newdata = onesample, type = "response")

  m1 <- onesample$meanY[onesample$interv == 1]
  m0 <- onesample$meanY[onesample$interv == 0]

  risk1 <- mean(m1)
  risk0 <- mean(m0)
```

```r
  rd <- risk1 - risk0
  rr <- risk1 / risk0

  c(risk1 = risk1, risk0 = risk0, rd = rd, rr = rr)
}

# Point estimates
est <- gcomp_est(nodes)
gcomp_rd <- est[3]
gcomp_rr <- est[4]

# Bootstrap CIs
set.seed(123)
B <- 1000 #Should be at least 1000 replicates for a bootstrap confidence interval

#Note: we bootstrap individuals here ignoring the network
#There are some instances that the model does not converge
boot_out <- replicate(B, {
  idx <- sample(seq_len(nrow(nodes)), replace = TRUE)
  gcomp_est(nodes[idx, , drop = FALSE])
})

boot_out <- t(boot_out)

ci <- apply(boot_out, 2, quantile, probs = c(0.025, 0.975), na.rm = TRUE)

results <- data.frame(
  measure = c("Risk under A=1", "Risk under A=0", "Risk difference", "Risk ratio"),
  estimate = est[c("risk1", "risk0", "rd", "rr")],
  lower_95_ci = ci[1, c("risk1", "risk0", "rd", "rr")],
  upper_95_ci = ci[2, c("risk1", "risk0", "rd", "rr")]
)

results$estimate <- round(results$estimate, 3)
results$lower_95_ci <- round(results$lower_95_ci, 3)
results$upper_95_ci <- round(results$upper_95_ci, 3)

results
```

```
##                measure estimate lower_95_ci upper_95_ci
## risk1  Risk under A=1    0.487       0.319       0.658
## risk0  Risk under A=0    0.486       0.417       0.556
## rd    Risk difference    0.002      -0.179       0.184
## rr         Risk ratio    1.003       0.649       1.422
```

```r
#save confidence intervals
gcomp_rd_ci <- c(results$lower_95_ci[1],results$upper_95_ci[1])
gcomp_rr_ci <- c(results$lower_95_ci[2],results$upper_95_ci[2])
```

Estimation using an Augmented IPW Estimator (doubly-robust). Two estimators below - augmented IPW (DR1, Lunceford and Davidian 2004) and weighted regression estimator (DR2, Robins 2007).

```
##################################################
# 7. Doubly Robust Estimators
##################################################

# Outcome model for g-formula components used in DR1
nodes_obs <- nodes
nodes_obs$interv <- -1

nodes_0 <- nodes
nodes_0$interv <- 0
nodes_0$treatment <- 0
nodes_0$outcome <- NA

nodes_1 <- nodes
nodes_1$interv <- 1
nodes_1$treatment <- 1
nodes_1$outcome <- NA

stacked <- rbind(nodes_obs, nodes_0, nodes_1)

outcome_mod <- glm(
  outcome ~ treatment + age10 + posneg + date + share + education + employment,
  data = stacked,
  family = binomial(link = "logit")
)

stacked$mean_outcome <- predict(outcome_mod, newdata = stacked, type = "response")

nodes$m0 <- stacked$mean_outcome[stacked$interv == 0]
nodes$m1 <- stacked$mean_outcome[stacked$interv == 1]
nodes$denom.p <- denom.p

# DR1: augmented IPW / Lunceford-Davidian
dr1_risk1 <- mean(
  (nodes$treatment * nodes$outcome - (nodes$treatment - nodes$denom.p) * nodes$m1) / nodes$denom.p
)

dr1_risk0 <- mean(
  ((1 - nodes$treatment) * nodes$outcome + (nodes$treatment - nodes$denom.p) * nodes$m0) / (1 - nodes$de
)

dr1_rd <- dr1_risk1 - dr1_risk0
dr1_rr <- dr1_risk1 / dr1_risk0

# DR2: weighted regression / Robins
nodes$ipw <- with(nodes, treatment / denom.p + (1 - treatment) / (1 - denom.p))

dr2_rd_mod <- lm(
  outcome ~ treatment + age10 + posneg +date + share + education + employment,
  data = nodes,
```

```r
  weights = ipw
)

dr2_rr_mod <- glm(
  outcome ~ treatment + age10 + posneg +date + share + education + employment,
  data = nodes,
  weights = ipw,
  family = binomial(link = "logit")
)
```

```
## Warning in eval(family$initialize): non-integer #successes in a binomial glm!
```

```r
#Warning in eval(family$initialize) :
  #non-integer #successes in a binomial glm!
  #Warning OK because we weighted glm

# Causal risk difference

nodes_1_dr <- nodes
nodes_1_dr$treatment <- 1

nodes_0_dr <- nodes
nodes_0_dr$treatment <- 0

dr2_rd_m1 <- predict(dr2_rd_mod, newdata = nodes_1_dr, type = "response")
dr2_rd_m0 <- predict(dr2_rd_mod, newdata = nodes_0_dr, type = "response")
dr2_rd <- mean(dr2_rd_m1) - mean(dr2_rd_m0)

dr2_rr_m1 <- predict(dr2_rr_mod, newdata = nodes_1_dr, type = "response")
dr2_rr_m0 <- predict(dr2_rr_mod, newdata = nodes_0_dr, type = "response")
dr2_risk1 <- mean(dr2_rr_m1)
dr2_risk0 <- mean(dr2_rr_m0)
dr2_rr <- dr2_risk1 / dr2_risk0

dr_results <- data.frame(
  estimator = c("DR1", "DR1", "DR2", "DR2"),
  measure = c("Risk difference", "Risk ratio", "Risk difference", "Risk ratio"),
  estimate = c(dr1_rd, dr1_rr, dr2_rd, dr2_rr)
)

dr_results
```

```
##   estimator         measure     estimate
## 1       DR1 Risk difference 0.0006497218
## 2       DR1      Risk ratio 1.0013315917
## 3       DR2 Risk difference 0.0083692436
## 4       DR2      Risk ratio 1.0294598410
```

```r
#save results

dr1_rd <- dr_results$estimate[1]
dr1_rr <- dr_results$estimate[2]
dr2_rd <- dr_results$estimate[3]
dr2_rr <- dr_results$estimate[4]
```

```
##################################################
# 8. Doubly Robust Estimators - Bootstrap CIs
##################################################
# Under SUTVA and IID, we sample each individual with replacement for the boot samples
#Turn into a function for DR1 and DR2 to bootstrap to get 95% CIs

estimate_effects <- function(data) {
  # ----------------------------
  # G-computation
  # ----------------------------
  dat_obs <- data
  dat_obs$interv <- -1

  dat_0 <- data
  dat_0$interv <- 0
  dat_0$treatment <- 0
  dat_0$outcome <- NA

  dat_1 <- data
  dat_1$interv <- 1
  dat_1$treatment <- 1
  dat_1$outcome <- NA

  stacked <- rbind(dat_obs, dat_0, dat_1)


withCallingHandlers(
  {
    outcome_mod <- glm(
    outcome ~ treatment + age10 + date + posneg + share + education + employment,
    data = stacked,
    family = binomial(link = "logit")
        )
  },
  #suppress warning for weighting in the glm
  warning = function(w) {
    if (grepl("non-integer #successes in a binomial glm!", conditionMessage(w))) {
      invokeRestart("muffleWarning")
    }
  }
)

  stacked$mean_outcome <- predict(outcome_mod, newdata = stacked, type = "response")

  m1 <- stacked$mean_outcome[stacked$interv == 1]
  m0 <- stacked$mean_outcome[stacked$interv == 0]

  g_risk1 <- mean(m1)
  g_risk0 <- mean(m0)
  g_rd <- g_risk1 - g_risk0
  g_rr <- g_risk1 / g_risk0

  data$m1 <- m1
```

```r
  data$m0 <- m0

  # -----------------------------
  # DR1: augmented IPW
  # w is the propensity score P(treatment = 1 | L)
  # -----------------------------
  dr1_risk1 <- mean(
    (data$treatment * data$outcome - (data$treatment - data$denom.p) * data$m1) / data$denom.p
  )

  dr1_risk0 <- mean(
    ((1 - data$treatment) * data$outcome + (data$treatment - data$denom.p) * data$m0) / (1 - data$denom
  )

  dr1_rd <- dr1_risk1 - dr1_risk0
  dr1_rr <- dr1_risk1 / dr1_risk0

  # -----------------------------
  # DR2: weighted regression
  # -----------------------------
  ipw <- data$treatment / data$denom.p + (1 - data$treatment) / (1 - data$denom.p)
  data$ipw <- ipw

  wreg_rd <- lm(
    outcome ~ treatment + age10 + date + posneg + share + education + employment,
    data = data,
    weights = ipw
  )

  withCallingHandlers(
  {
    wreg_rr <- glm(
    outcome ~ treatment + age10 + date + posneg + share + education + employment,
    data = data,
    weights = ipw,
    family = binomial(link = "logit")
        )
  },
  #suppress warning for weighting in the glm
  warning = function(w) {
    if (grepl("non-integer #successes in a binomial glm!", conditionMessage(w))) {
      invokeRestart("muffleWarning")
    }
  }
)

  dat1 <- data
  dat1$treatment <- 1

  dat0 <- data
  dat0$treatment <- 0

  rd_m1 <- predict(wreg_rd, newdata = dat1, type = "response")
```

```r
  rd_m0 <- predict(wreg_rd, newdata = dat0, type = "response")
  dr2_rd <- mean(rd_m1) - mean(rd_m0)

  rr_m1 <- predict(wreg_rr, newdata = dat1, type = "response")
  rr_m0 <- predict(wreg_rr, newdata = dat0, type = "response")
  dr2_risk1 <- mean(rr_m1)
  dr2_risk0 <- mean(rr_m0)
  dr2_rr <- dr2_risk1 / dr2_risk0

  c(
    gcomp_rd = g_rd,
    gcomp_rr = g_rr,
    dr1_rd = dr1_rd,
    dr1_rr = dr1_rr,
    dr2_rd = dr2_rd,
    dr2_rr = dr2_rr
  )
}

set.seed(123)
B <- 1000

point_est <- estimate_effects(nodes)

boot_est <- replicate(B, {
  idx <- sample(seq_len(nrow(nodes)), replace = TRUE)
  estimate_effects(nodes[idx, , drop = FALSE])
})


boot_est <- t(boot_est)

ci <- apply(boot_est, 2, quantile, probs = c(0.025, 0.975), na.rm = TRUE)

results_dr <- data.frame(
  estimator = c("G-computation", "G-computation", "DR1", "DR1", "DR2", "DR2"),
  measure = c("Risk difference", "Risk ratio",
              "Risk difference", "Risk ratio",
              "Risk difference", "Risk ratio"),
  estimate = point_est[c("gcomp_rd", "gcomp_rr", "dr1_rd", "dr1_rr", "dr2_rd", "dr2_rr")],
  lower_95_ci = ci[1, c("gcomp_rd", "gcomp_rr", "dr1_rd", "dr1_rr", "dr2_rd", "dr2_rr")],
  upper_95_ci = ci[2, c("gcomp_rd", "gcomp_rr", "dr1_rd", "dr1_rr", "dr2_rd", "dr2_rr")]
)

results_dr$estimate <- round(results_dr$estimate, 3)
results_dr$lower_95_ci <- round(results_dr$lower_95_ci, 3)
results_dr$upper_95_ci <- round(results_dr$upper_95_ci, 3)

results_dr
```

```
##              estimator         measure estimate lower_95_ci upper_95_ci
## gcomp_rd G-computation Risk difference    0.002      -0.179       0.184
## gcomp_rr G-computation      Risk ratio    1.003       0.649       1.422
## dr1_rd             DR1 Risk difference    0.001      -0.189       0.187
```

```
## dr1_rr              DR1      Risk ratio    1.001       0.634       1.424
## dr2_rd              DR2 Risk difference    0.008      -0.170       0.186
## dr2_rr              DR2      Risk ratio    1.029       0.670       1.418
```

```r
dr1_rd_ci <- c(results_dr$lower_95_ci[3],results_dr$upper_95_ci[3])
dr1_rr_ci <- c(results_dr$lower_95_ci[4],results_dr$upper_95_ci[4])
dr2_rd_ci <- c(results_dr$lower_95_ci[5],results_dr$upper_95_ci[5])
dr2_rr_ci <- c(results_dr$lower_95_ci[6],results_dr$upper_95_ci[6])

# Suppressed: Warning in eval(family$initialize): non-integer #successes in a binomial glm!
```

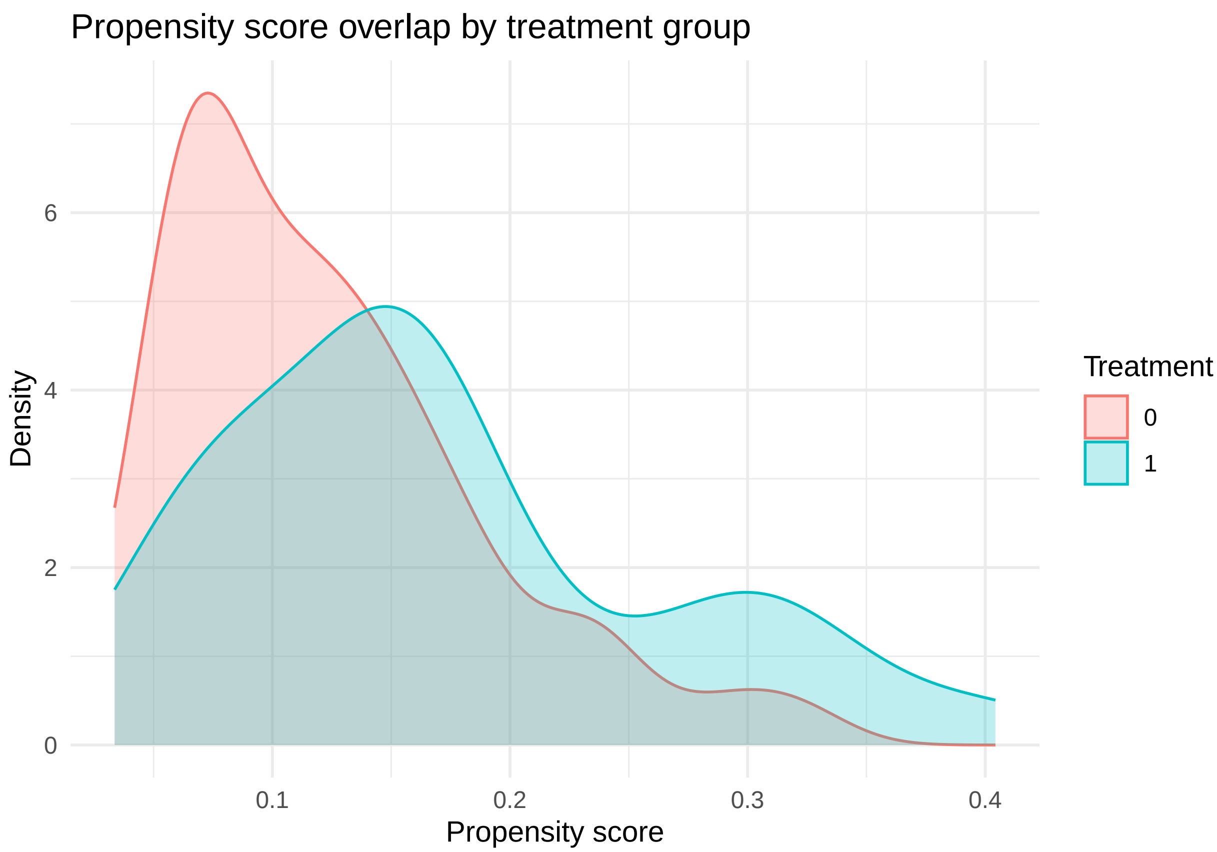
```r
#Code to output results to Latex (optional)
# z <- qnorm(0.975)
#
# make_rr_row <- function(estimator, fit, term = "treatment") {
#   b <- coef(fit)[term]
#   se <- sqrt(vcov(fit)[term, term])
#   est <- exp(b)
#   ci <- exp(b + c(-1, 1) * z * se)
#
#   data.frame(
#     Estimator = estimator,
#     Measure = "Risk ratio",
#     Estimate = est,
#     LCL = ci[1],
#     UCL = ci[2]
#   )
# }
#
# make_rd_row <- function(estimator, fit, term = "treatment") {
#   b <- coef(fit)[term]
#   se <- sqrt(vcov(fit)[term, term])
#   ci <- b + c(-1, 1) * z * se
#
#   data.frame(
#     Estimator = estimator,
#     Measure = "Risk difference",
#     Estimate = b,
#     LCL = ci[1],
#     UCL = ci[2]
#   )
# }
#
# make_manual_row <- function(estimator, measure, est, ci) {
#   data.frame(
#     Estimator = estimator,
#     Measure = measure,
#     Estimate = est,
#     LCL = ci[1],
#     UCL = ci[2]
#   )
# }
#
# results <- bind_rows(
#   make_rr_row("Unadjusted", model_rr_unadj),
```

```
#   make_rd_row("Unadjusted", model_rd_unadj),
#
#   make_rr_row("Covariate adjusted", gee_rr),
#   make_rd_row("Covariate adjusted", gee_rd),
#
#   make_rr_row("IPW", gee_rr_ipw),
#   make_rd_row("IPW", gee_rd_ipw),
#
#   make_manual_row("G-computation", "Risk ratio", gcomp_rr, gcomp_rr_ci),
#   make_manual_row("G-computation", "Risk difference", gcomp_rd, gcomp_rd_ci),
#
#   make_manual_row("DR1", "Risk ratio", dr1_rr, dr1_rr_ci),
#   make_manual_row("DR1", "Risk difference", dr1_rd, dr1_rd_ci),
#
#   make_manual_row("DR2", "Risk ratio", dr2_rr, dr2_rr_ci),
#   make_manual_row("DR2", "Risk difference", dr2_rd, dr2_rd_ci)
# )
#
#
# table_wide <- results %>%
#   mutate(
#     Estimate = round(Estimate, 3),
#     LCL = round(LCL, 3),
#     UCL = round(UCL, 3),
#     value = sprintf("%.3f (%.3f, %.3f)", Estimate, LCL, UCL)
#   ) %>%
#   select(Estimator, Measure, value) %>%
#   pivot_wider(names_from = Measure, values_from = value) %>%
#   select(Estimator, `Risk ratio`, `Risk difference`)
#
# latex_tab <- kable(
#   table_wide,
#   format = "latex",
#   booktabs = TRUE,
#   escape = FALSE,
#   align = c("l", "l", "l")
# )
#
# slide_tex <- paste0(
#   "\\begin{frame}{Effect Estimates}\n",
#   "\\scriptsize\n",
#   "\\centering\n",
#   latex_tab,
#   "\n\\end{frame}\n"
# )
#
# writeLines(slide_tex, "effect_estimates_slide.tex")
```

Assess the fit of the propensity score model using overlap plots and standardized mean differences.

```
##################################################
# 9. Assess the propensity score model
```

```r
###################################################

# IPW from propensity score
nodes$ipw <- with(nodes, treatment / denom.p + (1 - treatment) / (1 - denom.p))

# Density plot of propensity scores by treatment group
ggplot(nodes, aes(x = denom.p, color = factor(treatment), fill = factor(treatment))) +
  geom_density(alpha = 0.25) +
  labs(
    x = "Propensity score",
    y = "Density",
    color = "Treatment",
    fill = "Treatment",
    title = "Propensity score overlap by treatment group"
  ) +
  theme_minimal()
```




```r
#Standardized Mean Differences
bal <- bal.tab(
  treatment ~ age10 + date + posneg + share + education + employment,
  data = nodes,
  weights = nodes$ipw,
  un = TRUE,
  method = "weighting",
  estimand = "ATE"
)
```

```r
print(bal)
```

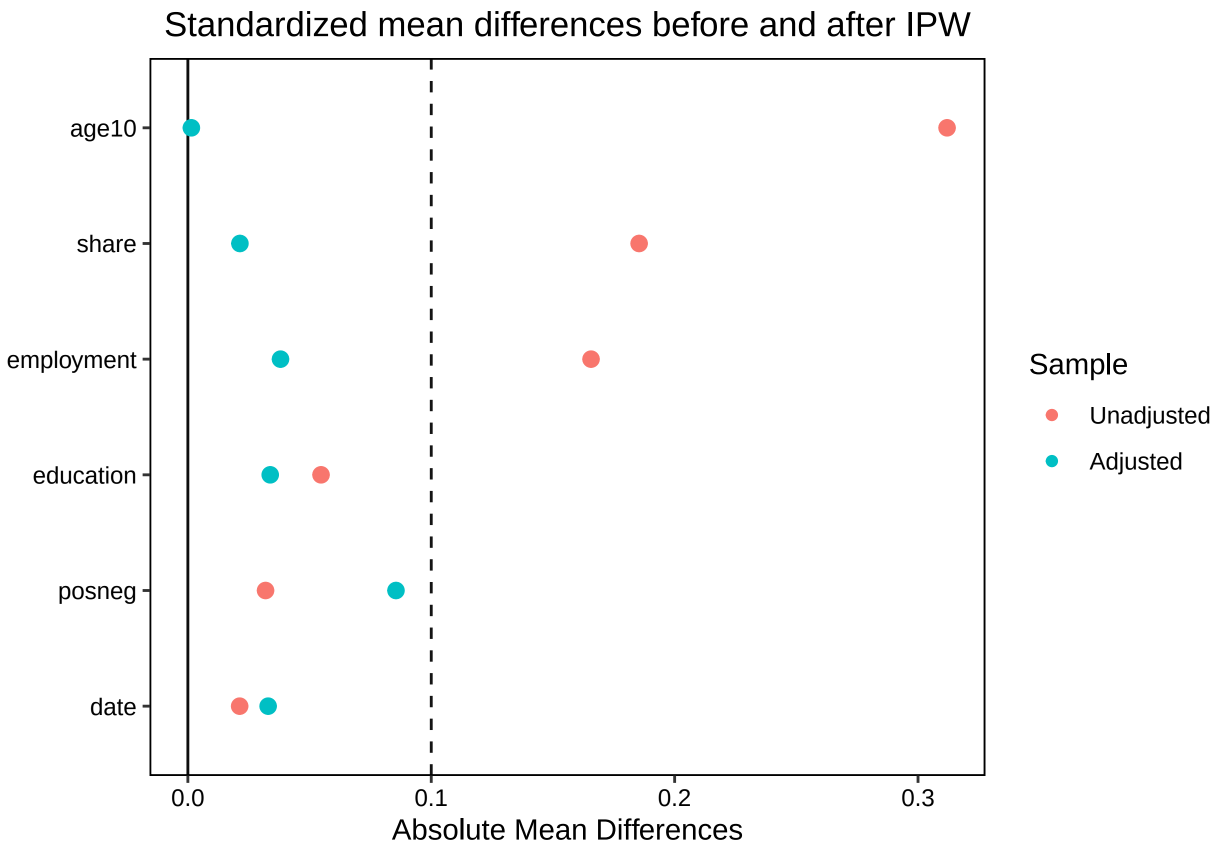
```
## Balance Measures
##                Type Diff.Un Diff.Adj
## age10       Contin. -0.3119  -0.0014
## date         Binary -0.0213  -0.0330
## posneg       Binary -0.0319   0.0855
## share        Binary -0.1854   0.0214
## education    Binary  0.0547   0.0338
## employment   Binary  0.1657  -0.0381
##
## Effective sample sizes
##            Control Treated
## Unadjusted  188.     28.
## Adjusted    186.75   20.64
```

```r
smd_table <- bal$Balance
smd_table[, c("Diff.Un", "Diff.Adj")]
```

```
##                Diff.Un     Diff.Adj
## age10      -0.31192909 -0.001435053
## date       -0.02127660 -0.032982923
## posneg     -0.03191489  0.085505547
## share      -0.18541033  0.021381623
## education   0.05471125  0.033832176
## employment  0.16565350 -0.038061197
```

```r
#Love Plot

love.plot(
  bal,
  stat = "mean.diffs",
  abs = TRUE,
  thresholds = c(m = 0.1),
  var.order = "unadjusted",
  title = "Standardized mean differences before and after IPW"
)
```

```
## Warning: Standardized mean differences and raw mean differences are present in the same
## plot. Use the `stars` argument to distinguish between them and appropriately
## label the x-axis. See `love.plot()` for details.
```



Standardized mean differences before and after IPW
age10
share
employment
education
posneg
date
0.0
0.1
0.2
0.3
Absolute Mean Differences
Sample
Unadjusted
Adjusted